

Win Nodes for Bayes Nets: Exact Order-Statistic Queries on Gaussian Graphical Models

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Abstract

The inference stack of a Gaussian graphical model returns marginals, joint moments and samples, and none of these answers the questions a user of a correlated field most often asks: which component is the maximum, when is a threshold first crossed, who wins. We formalize these as the queries of a *win node*, an n -ary node $W = \arg \min_i X_i$ adjoined to Gaussian variables, and observe that for factor-structured and banded-precision fields the node’s forward query, its derivatives, and its use as an observation are all exact and computable in time linear in n . The queries are not functions of the marginals: a stationary AR(1) chain has identical marginals at every index while its maximum is $2.6\times$ likelier to sit at an endpoint than in the middle, a fact the win node computes in milliseconds and Monte Carlo confirms.

1 Queries that are not marginal functionals

Let $X = (X_1, \dots, X_n)$ be jointly Gaussian: the setting of the Kalman filter, the Gaussian Markov random field, and the Gaussian belief-propagation stack. Every mainstream toolkit returns marginal moments, covariances, samples, and evidence. Four common queries are functionals of the joint law that no collection of low-dimensional marginals determines.

1. Argmax marginals $a_t = \Pr(X_t = \max_s X_s)$: which sensor reads the extreme, which index carries the peak, which entrant wins.
2. The law of the maximum $F_M(u) = \Pr(\max_t X_t \leq u)$, whose complement is the excursion probability of spatial statistics.
3. First passage, the distribution of $\tau = \min\{t : X_t > u\}$: sequential monitoring, degradation to failure, barrier crossing.
4. Top- k membership $\Pr(X_i \text{ among the } k \text{ smallest})$: place and show, portfolio inclusion, admission.

The stationary AR(1) chain makes the gap concrete. Take $X_{t+1} = \phi X_t + \sqrt{1 - \phi^2} \epsilon_t$ with $X_1 \sim N(0, 1)$: every marginal is $N(0, 1)$, and a stationary Kalman filter reports exactly that at every index. The argmax marginals at $\phi = 0.9$, $n = 20$, computed by the method of Section 5 and verified against 4×10^5 -path Monte Carlo to 0.007, put mass 0.091 on each endpoint and 0.035 in the middle. Interior points are squeezed by correlated-high neighbours on both sides;

endpoints have one neighbour. The fact lives in the joint law, so the marginal layer cannot state it.

We call the object that answers these queries a *win node*: a deterministic n -ary node $W = \arg \min_i X_i$ (min-wins; negate for utilities) adjoined to the Gaussian variables, together with its order-statistic relatives. The definition is trivial. The observation of this paper is that for the two coupling structures that dominate practice, the node’s forward query, its derivatives, and its conditioning rule are exact and cheap at the same time, which is what a graphical-model query type requires.

The ingredients are classical, and we say so once. The one-dimensional winning-time integral is the random-utility representation of [Domencich and McFadden \(1975\)](#); the shared survival field is competing-risks calculus; the transfer operator on a Markov chain is the sum-product algorithm; the derivative identity of Section 3 was reached for general joint laws by [Müller et al. \(2019\)](#). What is new is the organization: one field pass per query for all n components, linear-work derivatives, and the chain and band order-statistic layer with its boundary corrections, all measured below.

2 Adjacent machinery

TrueSkill ([Herbrich et al., 2007](#)) is a Bayes net with win nodes already: its factor graph attaches greater-than factors to performance differences and propagates moment-matched messages. Its win node is binary, multi-entrant events are chains of pairwise approximations, and the factor itself is approximated by expectation propagation. Section 4 gives the exact n -ary likelihood message; the TrueSkill factor is its $n = 2$ case.

Modern Gaussian message-passing systems and GMRF libraries compute marginals, free energies and samples for very large models. Their query vocabularies do not include the four functionals above. The win node is proposed as an addition to that vocabulary, not as a competitor to the toolkits.

For latent Gaussian models, [Bolin and Lindgren \(2015\)](#) compute excursion sets and exceedance probabilities by sequential importance sampling, and their `excursions` package is the standard tool. On chains and bands the same functionals admit the deterministic pass of Section 5. Simulation remains the recourse for general sparsity, and our software points there at its refusal boundary.

GHK evaluates one orthant probability per alternative by sequential importance sampling. [Ridgway \(2016\)](#) showed that on an AR(1) covariance its normalized variance grows exponentially in dimension. That is the correlated-chain case where the restricted transfer operator is exact and linear in n : high correlation is where the exact pass is tightest and the simulator is worst, and Section 6 measures the former.

3 The win node on factor-structured fields

Let $X = \mu + Vf + \text{diag}(\sqrt{D})\epsilon$ with $f \sim N(0, I_r)$ and independent ϵ . Conditional on f the components are independent. With F_j , $S_j = 1 - F_j$ and f_j the conditional CDF, survival and density of X_j ,

$$a_i = \Pr(X_i = \min_j X_j) = E_f \int f_i(x) \prod_{j \neq i} S_j(x) dx. \quad (1)$$

In English: component i wins by finishing at some value x with everyone else still above it, averaged over the shared luck.

The product is shared. Build the log survival field $L(x) = \sum_j \log S_j(x)$ once on a lattice of L points; each leave-one-out product is the subtraction $e^{L - \log S_i}$. One pass prices all n win probabilities at $O(nL)$ per factor node, so $O(nLQ)$ for a Q -node quadrature over f . Top- k membership replaces the survival product with a Poisson–binomial count program over the F_j , deconvolved per component in the stable direction, at the same cost profile.

Derivatives come from the divergence theorem. Component i wins on the cone $R_i = \{x : x_i \leq x_k \ \forall k\}$, and translating μ_j moves probability across the shared face F_{ij} only, so for $j \neq i$

$$\frac{\partial a_i}{\partial \mu_j} = \frac{1}{\sqrt{2}} \int_{F_{ij}} q_\mu dS \geq 0, \quad \frac{\partial a_i}{\partial \mu_i} = - \sum_{j \neq i} \frac{\partial a_i}{\partial \mu_j}. \quad (2)$$

In English: the derivative is the tie density, the Jacobian is symmetric with zero row sums, and it is therefore minus a weighted graph Laplacian on the photo-finish graph. The weights factor through the field, $w_{ij} = \int G A_i A_j dx$ with hazards $A_i = f_i/S_i$ and $G = \prod_k S_k$, so the dense matrix is applied in $O(nL)$ without forming an edge. The count-program analogue yields both top- k Jacobians $(\partial q/\partial \mu, \partial q/\partial \sigma)$, checked by the scale-gauge Euler identity $\mu \cdot \partial_\mu q + \sigma \cdot \partial_\sigma q = 0$. Derivations and the inversion theory are in [Cotton \(2026\)](#). Here they make the node trainable: its output can be matched to observed win frequencies by Newton methods with exact gradients.

4 Conditioning on who won

A query node becomes a Bayes-net citizen when it can be observed. Condition on $W = k$ and integrate out the factor and the winner’s own noise z :

$$\Pr(W = k \mid \mu) = E_{f,z} \prod_{j \neq k} \Phi \left(\frac{\mu_k - \mu_j + (Vf)_k - (Vf)_j + z\sqrt{D_k}}{\sqrt{D_j}} \right). \quad (3)$$

In English: given the shared luck and the winner’s own performance, each rival independently had to finish behind, and each such event is one normal CDF.

The score of (3) is analytic: posterior node weights times Mills ratios, one extra pass over arrays the likelihood already computes. The message an observed win node passes to its Gaussian parents therefore carries exact first-order information. At $n = 2$ and $r = 0$ the likelihood reduces to $\Phi((\mu_k - \mu_j)/\sqrt{D_k + D_j})$, which is TrueSkill’s greater-than factor. The n -ary node needs no pairwise chaining and no moment approximation of the factor itself; moment approximations enter only if the user chooses a Gaussian posterior family, exactly as in expectation propagation. This likelihood and score are the estimation core of the multinomial-probit software below, validated against finite differences and against dual-number automatic differentiation at 10^{-10} .

5 The order-statistic layer on chains and bands

Now let the coupling be Markov with precision bandwidth b ; the Gauss–Markov chain is $b = 1$. Write the chain as $X_1 \sim N(\mu_1, \sigma_0^2)$ and $X_{t+1} \mid X_t \sim N(\mu_{t+1} + \phi_t(X_t - \mu_t), s_t^2)$. A tridiagonal precision matrix delivers these parameters directly: its Cholesky factor is bidiagonal, and each row is a conditional regression.

Max-CDF and first passage. Restrict the transfer operator below u . With T_t the transition kernel on the lattice and π_u the initial sub-density cut at u ,

$$\Pr(\max_{s \leq t} X_s \leq u) = \mathbf{1}^\top \left(\prod_{s < t} T_s^{(u)} \right) \pi_u, \quad T^{(u)}(x' | x) = T(x' | x) \mathbf{1}\{x' \leq u\}, \quad (4)$$

one forward pass, $O(nL)$ per threshold with banded kernels. The survivor-mass decrements of the same pass are the first-passage law. Integrating the survival of the max over a threshold grid gives $E[\max]$; the rectangle rule biases this by $O(\Delta u)$, measured at $+0.053$ on a ten-node chain, and the trapezoid removes it. A hard cell cutoff at u costs $O(\Delta x)$; assigning the boundary cell its fractional occupancy restores $O(\Delta x^2)$, which is the difference between 10^{-2} and 2×10^{-4} agreement with the i.i.d. closed form at 400 lattice points.

Argmax marginals. For each lattice threshold u_k , a forward pass restricted strictly below u_k and reaching $X_t = u_k$ gives $\alpha_t(k) = \Pr(X_1, \dots, X_{t-1} < u_k, X_t = u_k)$, and a backward conditional pass gives $\beta_t(k) = \Pr(X_{t+1}, \dots, X_n < u_k | X_t = u_k)$. Then

$$\Pr(X_t = \max_s X_s) = \sum_k \alpha_t(k) \beta_t(k), \quad (5)$$

integrating out the value of the maximum. With transitions precomputed the double pass is matrix slices, $O(nL^2)$ in total.

Bandwidth b by state lifting. The vector $(X_t, \dots, X_{t+b-1})$ is Markov, so the same restricted passes run on the lifted state at $O(nL^{b+1})$: linear in n , exponential only in the bandwidth. At $b = 2$ the lift covers AR(2), including the cyclical regime with complex roots that has no $b = 1$ analogue. With the second coefficient set to zero the lift reproduces the $b = 1$ pass to 5.6×10^{-16} on the same grid.

6 Numerical checks

All numbers come from committed, seeded scripts in the public repository. Monte Carlo appears only as a referee.

query	setting	exact vs. referee	cost
max-CDF	AR(1), $n=50$, $\phi=0.99$	5×10^{-3} (MC 4×10^5)	3 ms
max-CDF	i.i.d. closed form, $n=12$	2×10^{-4}	—
argmax	AR(1), $\phi=0.9$, $n=20$	7×10^{-3} (MC)	ms
argmax	random walk, $n=40$	4×10^{-3} (MC)	ms
band $b=2$	AR(2) incl. cyclical	1.1×10^{-2} (grid-limited)	230 ms/thr.
$b=2$ embed	$\phi_2=0$ vs. $b=1$	5.6×10^{-16}	—
first passage	inhomogeneous chain, $n=10$	5×10^{-3} (MC)	$O(nL)$
$E[\max]$	same	5×10^{-3} (MC band)	—

Two consequences follow from the table rather than merely illustrating it. First, the boundary pile-up law: positive serial correlation pushes the argmax of a stationary path to the endpoints, $2.6\times$ at $\phi = 0.9$, and no marginal summary can express this. Second, the finite- n arcsine refinement: on a random walk the transfer operator matches the finite- n argmax law to 0.3% while the continuum arcsine density sits 1% away, because the engine computes the exact finite

law of which arcsine is the limit. The same pass handles the drifted, barriered and mean-reverting cases that have no closed form. The accuracy profile is complementary to simulation: the exact pass is tightest at high correlation, exactly where [Ridgway \(2016\)](#) shows the GHK family degrading.

7 Software

The factor win node, its top- k relatives, both Jacobian blocks and the calibration inverses ship in the `winning` package: a Python reference with R, JavaScript, Julia and Rust ports pinned to embedded parity vectors, most at 10^{-15} . The observed win node, likelihood (3) with analytic score, is the estimation core of `MultinomialProbit.jl`, whose score is additionally refereed by forward-mode automatic differentiation through the same code path. The chain and band layer is `GMRFExtremes.jl`, which reads a Gauss–Markov chain off a tridiagonal precision and dispatches on GMRF library types. Wider bandwidth and general sparsity are refused with a pointer to the lifted pass and to [Bolin and Lindgren \(2015\)](#) respectively.

8 Scope

The reduction requires structure. Factor coupling of rank r costs a Q -node quadrature and is exact to quadrature accuracy; precision bandwidth b costs $O(nL^{b+1})$ and is practical for $b \leq 3$; general sparse precision gets neither treatment here, and the excursion functionals then belong to the sampling methods of [Bolin and Lindgren \(2015\)](#). The win-node concept itself is not new: TrueSkill carries a binary one, and [Müller et al. \(2019\)](#) derived the tie-density derivative for general laws. New here are the n -ary node as an exact, linear-work, differentiable query on factor and banded structures, the restricted-pass order-statistic layer with its boundary corrections and band lift, and the exact n -ary observed-win message. We leave to future work the lifted-state argmax at $b \geq 2$ and message compression past $b \approx 3$.

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